

Note 01 · Partition curves, and why Exercise 1 says Karra

d_{50} , sharpness, imperfection, efficiency; CYCB against SCR2; the numbers of the Kirunavaara screening station

1 The question this note answers

A screen never separates perfectly. Particles clearly larger than the holes stay on top, particles clearly smaller fall through, and the ones close to the hole size (the *near-size* particles) go one way or the other depending on how loaded the deck is, how long they travel on it and pure chance. So two things are needed: a way to describe what a real screen does with each size, and a way to choose, in MODSIM, between a model that *reproduces* a known separation and a model that *predicts* it from the machine. Exercise 1 asks for the second kind, and this note explains why.

2 A toy example before any theory

Imagine a single deck with 10 mm square holes and a feed of four sizes, 25 t/h each (100 t/h in total). Suppose we measured how much of each size ended up in the oversize:

Particle size (mm)	Feed (t/h)	To oversize (t/h)	Recovery to oversize
20	25	25.0	1.00
12	25	23.75	0.95
9	25	7.5	0.30
3	25	0.5	0.02
total	100	56.75	

The last column is the whole story: it is the **partition curve**. A perfect screen would show 1.00 for 20 and 12 mm and 0.00 for 9 and 3 mm. The real one leaks: 30 % of the 9 mm particles, which should have passed, stayed on top, and 5 % of the 12 mm particles slipped through (they never do in reality, unless the cloth is damaged, but a small value stands here for holes that are slightly larger than nominal).

Concept: Partition curve, d_{50} , sharpness, imperfection

The **partition curve** (also Tromp curve) gives, for each particle size, the fraction of the feed of that size that reports to the coarse product. It is read from data or computed by a model.

The **cut size** d_{50} is the size with a 50 % chance of going either way. In the toy example it sits between 9 mm (0.30) and 12 mm (0.95); interpolating on a straight line gives about 9.9 mm, a little *below* the 10 mm aperture. That is normal: a particle slightly smaller than the hole still needs a free hole under it at the right moment.

The **sharpness** says how steep the curve is around d_{50} . The course's formula sheet measures it with the **imperfection**

$$I = \frac{d_{75} - d_{25}}{2 d_{50}},$$

where d_{25} and d_{75} are the sizes with 25 % and 75 % recovery to oversize. In the toy example $d_{25} \approx 8.8$ mm and $d_{75} \approx 10.9$ mm, so $I \approx (10.9 - 8.8)/(2 \times 9.9) \approx 0.11$. A perfect step has $I = 0$; the larger I , the more near-size material goes the wrong way.

Concept: Screen efficiency, the formula-sheet way

The formula sheet defines

$$E = \frac{c - f}{c(1 - f)},$$

with f the fraction of *oversize material* (coarser than the cut) in the feed and c the same fraction in the coarse product. Toy numbers: the feed carries 50 t/h coarser than 10 mm out of 100, so $f = 0.50$; the coarse product carries $25 + 23.75 = 48.75$ t/h of true oversize out of 56.75, so $c = 0.859$. Then $E = (0.859 - 0.500)/(0.859 \times 0.500) = 0.84$. Read it as: the screen did 84 % of the separating job a perfect screen would have done.

3 The two ways MODSIM models a screen

MODSIM offers two families, and the choice is the heart of Exercise 1.

3.1 Give it the curve: the distribution-curve model (CYCB)

You type in the partition curve yourself, in practice a d_{50} and a sharpness. MODSIM applies that curve to whatever feed arrives. It is the right tool when you already *know* how an existing screen separates (from a plant survey, say) and you want the simulator to reproduce it. Its weakness is exactly its strength: the curve never changes. Feed the screen twice as much and it separates exactly the same way, which no real screen does.

3.2 Give it the machine: the Karra model (SCR2 single deck, DSC2 double deck)

You describe the physical screen: mesh (aperture) size, wire diameter, angle of inclination, deck length and width, bulk density of the material, screen type (woven wire or punched plate) and how many screens run in parallel. From that and from the feed the simulator supplies, the model computes a *basic capacity* in tonnes per hour per square metre, corrects it for the amount of near-size and oversize material, the open area and the other factors of the classical capacity methods, and derives the separation. The report gives the corrected capacity, the **area utilisation factor** (AUF: how loaded the deck is against what it can carry) and the resulting partition. Change the tonnage or the deck size and the answer changes, as it should.

There is also a simple ideal model (SCRN, DSC1: a mesh size plus a fixed transmission efficiency, 90 % by default) for a first sketch before any equipment is chosen.

In one sentence

CYCB reproduces a separation you already know; Karra predicts one from a machine you describe, so only Karra can answer *how big must the screen be* and *what happens if the tonnage changes*.

4 The numbers of the Kirunavaara station (Exercise 1)

The sheet gives 600 t/h of D-ore at 4.0 t/m³ and this cumulative size distribution (percent finer than the size):

Size (mm)	Cum % finer	Size (mm)	Cum % finer
600	99.5	30	49.9
300	95.1	15	36.9
100	84.3	10	30.0
70	73.5	5	19.8
50	64.1	3.3	15.1

Three fractions are wanted: coarser than 100 mm, 20 to 100 mm, and finer than 20 mm. Before opening MODSIM, the perfect-screen numbers:

- Coarser than 100 mm: $100 - 84.3 = 15.7\%$, that is $0.157 \times 600 = 94$ t/h.
- Finer than 20 mm: 20 mm is not in the table, so interpolate on a logarithmic size axis between 15 mm (36.9%) and 30 mm (49.9%): $\ln(20/15)/\ln(30/15) = 0.415$, hence $36.9 + 0.415 \times 13.0 = 42.3\%$, that is 254 t/h.
- Middle fraction: $84.3 - 42.3 = 42.0\%$, that is 252 t/h.

The coarse and the fines go together to the autogenous mills: $94 + 254 = 348$ t/h with $94/348 = 27\%$ coarser than 100 mm, comfortably above the 15% the sheet says a mill needs as grinding media. The middle fraction, 252 t/h, is what would become critical stones (20 to 50 mm) in the mill, so it goes to crushing.

These are upper bounds. Karra will move a few percent of near-size material to the oversize of each deck, so the coarse products come out a little heavier and slightly contaminated, the fine products a little lighter and cleaner. If the simulation lands far from 94/252/254, check the feed definition before anything else.

Watch out / typical mistake

The three things found in **Exercise 1.mpd** saved on 2 September, in order of damage:

1. both screens still at the default aperture of 0.0508 m (2 inches): the station cuts twice at 51 mm and never at 100 or 20 mm;
2. the size mesh stops at 0.01 m (10 mm) while the feed reaches 600 mm: 70% of the feed lands in the top class and cannot be screened at all;
3. the feed stream has the mineral and the density 4.0 but **feedRate 0** and **defined false**: the job runs without a single error message and every stream shows zeros.

Two more classics: the Swedish decimal comma (MODSIM reads 4,0 as garbage) and typing millimetres into a field the new GUI keeps in metres (100 mm is 0.100).

Try it yourself (keyboard in hand)

1. Open **Exercise 1.mpd** in the MODSIM-GUI (the dashboard button opens the GUI and selects the file in Explorer). Fix the three items above: apertures 0.100 and 0.020 m, top size 1.0 m with about 20 to 25 classes, feed 600 t/h with the ten sieve points.
2. Run. Read the fly-outs: do the three fractions land near 94, 252 and 254 t/h? Write down the differences; they are the near-size effect.
3. Open each screen report. Find the basic capacity, the corrected capacity and the AUF. If the AUF is far above 1, the deck is overloaded: make it larger and rerun; far below 1, it is oversized.
4. From the partition curve in the report, read d_{25} , d_{50} and d_{75} of the 100 mm deck and compute I . Then double the feed to 1200 t/h and recompute: with Karra, I gets worse; with CYCB it would not move at all. That difference is the whole reason the sheet says Karra.

5. Compare with **CobbScreen.PAK** and the suggested-solution workbook, both in the module's material folder, after your own run.

Glossary: in plain words

Aperture the size of the hole in the deck; the nominal cut.

Near-size particles between roughly 0.75 and 1.25 times the aperture; the ones a screen struggles with.

Oversize, undersize what stays on the deck and what passes through it.

Partition curve recovery to the coarse product as a function of size; Tromp curve is the same thing.

Deck one screening surface; a double-deck screen stacks two with different apertures.

AUF area utilisation factor, how loaded the deck is compared with its corrected capacity.

Fly-out MODSIM's small box beside a stream showing its solids flow, water flow and percent solids; the exercise wants one on every stream.